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GWCopyPro — User Handbook

Version 1.0 · For GWCopyPro with gw.exe v0.24+ · by The8BitBox™ — Ilija Injac



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1. About this Handbook

This handbook explains every function of **GWCopyPro** in detail. It is written so that newcomers who have never used a GreaseWeazle before can follow along: every technical term is explained in the [Glossary](#), and every command-line flag of the underlying `gw.exe` tool is described in plain language in the [gw.exe Parameter Dictionary](#).

2. Introduction — GreaseWeazle and GWCopyPro

2.1 What is a GreaseWeazle?

A [GreaseWeazle](#) is a small, inexpensive open-source USB device designed by Keir Fraser. You connect it between your PC (via USB) and an ordinary floppy disk drive (via the classic 34-pin floppy ribbon cable). Unlike a normal USB floppy drive, the GreaseWeazle does not care about *file systems* or *formats* at all — it records the **raw magnetic flux** on the disk surface, i.e. the exact stream of magnetic reversals that the drive head sees.

Because of this, a GreaseWeazle can read and write **virtually any floppy format ever made**: IBM PC, Amiga, Atari ST, Atari 8-bit, Commodore 64/128, Apple II, Macintosh, MSX, PC-98, Acorn, and many more — including copy-protected disks, as long as the connected drive is mechanically compatible (3.5", 5.25", or even 8").

The official software for the GreaseWeazle is a **command-line tool** called `gw.exe` (the "Greaseweazle host tools"). It is powerful, but it must be operated by typing commands such as:

```
gw read --device COM3 --format ibm.1440 --tracks=c=0-79:h=0-1 --retries 3
mydisk.img
```

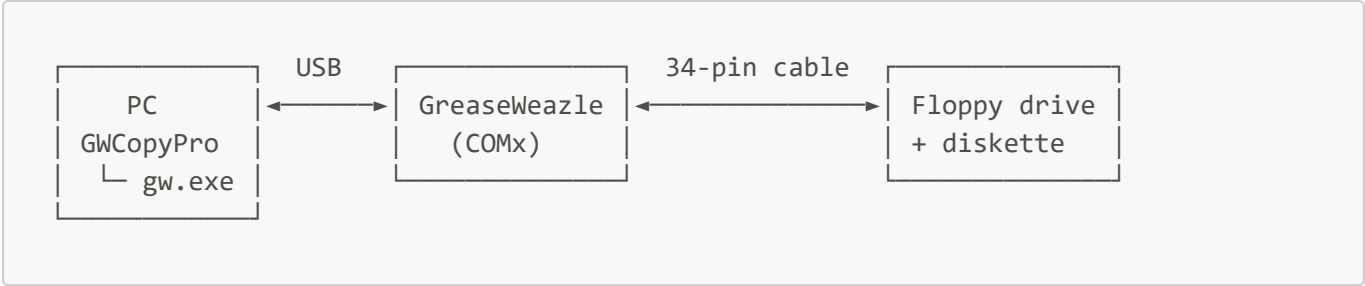
2.2 What is GWCopyPro?

GWCopyPro is a graphical Windows application that wraps `gw.exe` in a comfortable, dark-themed user interface. It builds those command lines for you — you click checkboxes and pick values, and GWCopyPro shows you the exact `gw.exe` command it will run.

Its highlights:

- **Multiple GreaseWeazle devices** at once — each on its own COM port, each usable in parallel.
- **Read and write jobs** with a live, track-by-track colour visualiser.
- **Repetitive mode** for digitising whole boxes of disks: insert, image, swap, repeat — with automatic file numbering.
- **Post-actions**: run programs or scripts automatically after each successful job (checksums, archiving, extraction, conversion...).
- **Presets**: save any job configuration and reload it with two clicks.
- **Detailed logs** for every job.
- English and German user interface.

2.3 How the pieces fit together



GWCopyPro never talks to the hardware directly — it always launches `gw.exe`, feeds it the parameters you chose, and interprets its output live.

3. Requirements and Installation

3.1 What you need

Component	Details
Windows 10 / 11	64-bit recommended
.NET 8.0 Desktop Runtime	Required to run GWCopyPro
<code>gw.exe v0.24 or newer</code>	From the official GreaseWeazle tools package
One or more GreaseWeazle devices	Any model (V4, V4.1, F1, F7, ...)
A floppy drive	3.5", 5.25", or 8" — with power supply and 34-pin data cable

Important: GWCopyPro generates the `--tracks=` compound syntax introduced in `gw.exe v0.24`. Older versions of `gw.exe` that still use `--scyl` / `--ecyl` will **not** work with the track selection features.

3.2 Installing gw.exe

1. Download the latest *Greaseweazle host tools* ZIP from github.com/keirf/greaseweazle/releases.
2. Unpack it to a folder of your choice, e.g. `C:\gw\`.
3. Either add that folder to your Windows `PATH`, **or** point GWCopyPro to the full path of `gw.exe` in **Settings** (see [chapter 11](#)).

When you plug in a GreaseWeazle for the first time, Windows creates a virtual **COM port** for it (e.g. `COM3`). You can check which port was assigned in the Windows *Device Manager* under *Ports (COM & LPT)* — but normally GWCopyPro's auto-detection finds it for you.

3.3 Installing GWCopyPro

Unpack (or build) GWCopyPro into any folder. The folder contains:

<code>GWCopyPro.exe</code>	the application	
<code>icon\favicon.ico</code>	application icon	
<code>tools\lsar.exe</code>	bundled archive lister	(The Unarchiver command-line tools)

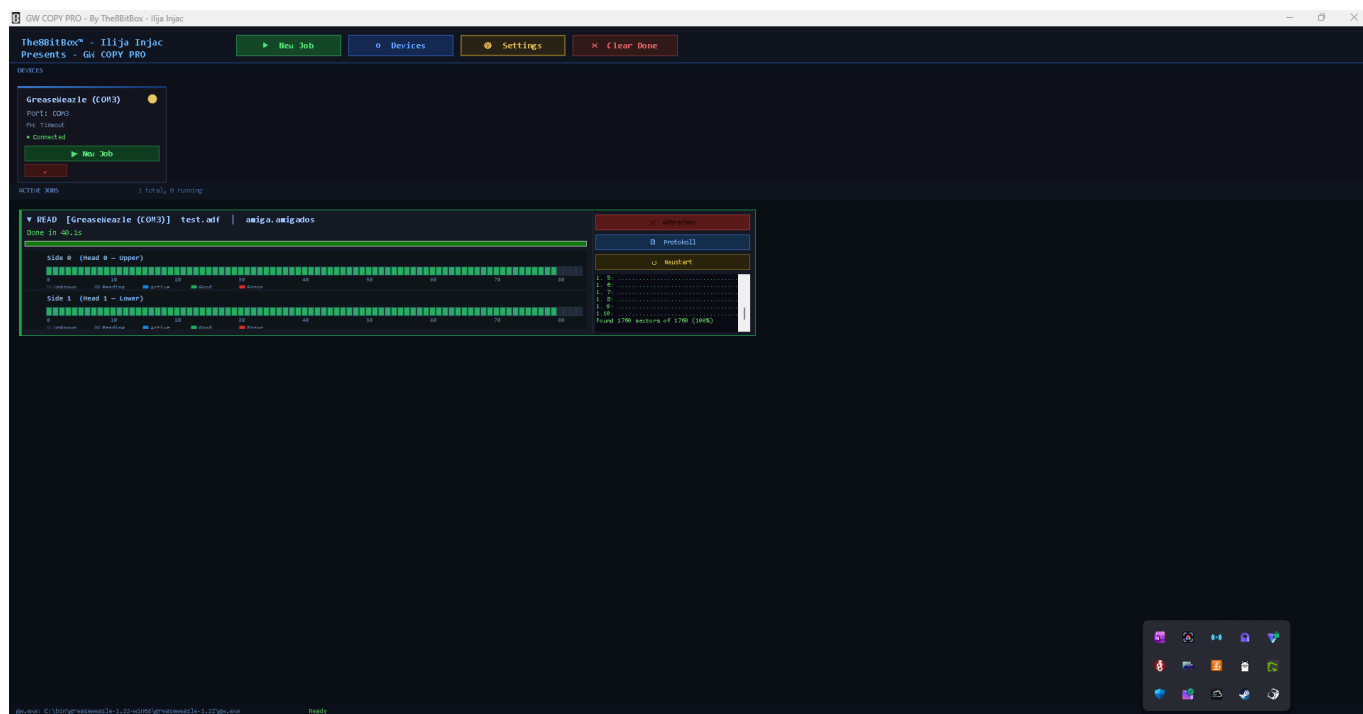
tools\unar.exe	bundled archive extractor
Logs\	created automatically – one subfolder per job

All user data lives under `%APPDATA%\GreaseWeazleManager\`:

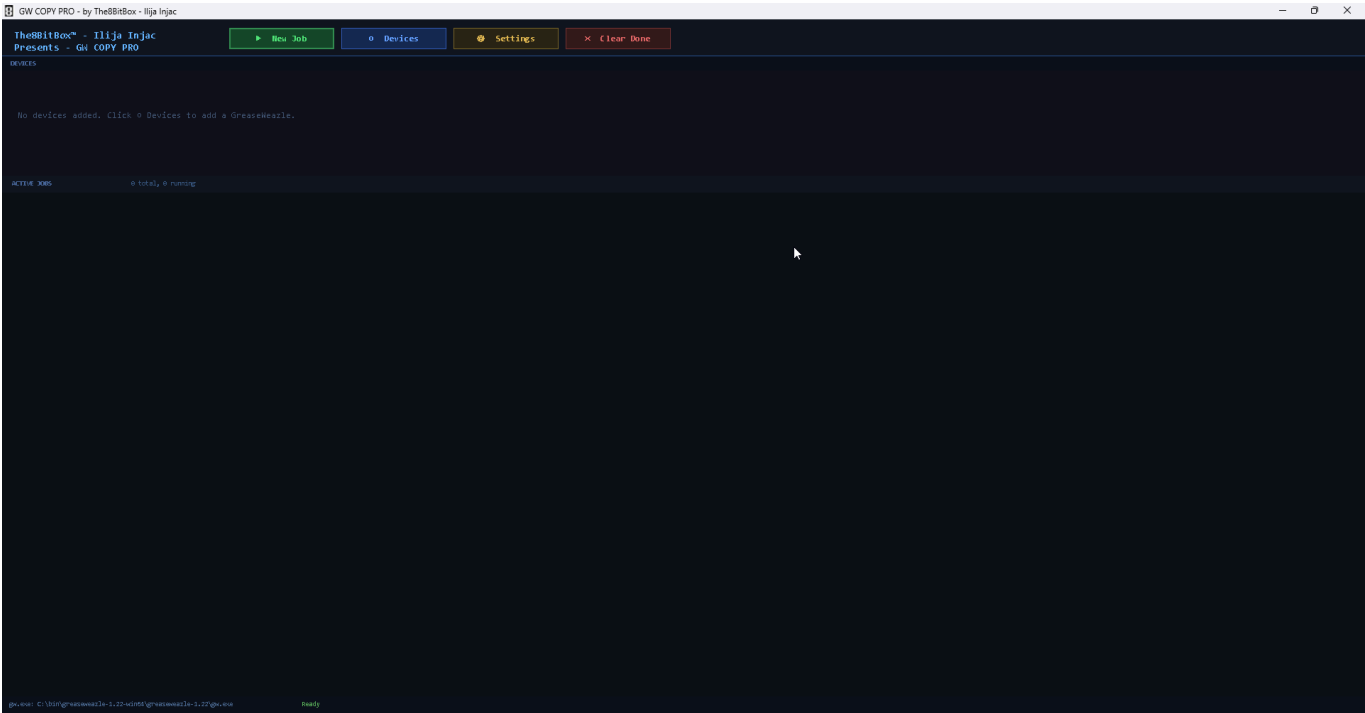
<code>%APPDATA%\GreaseWeazleManager\</code>	
settings.json	application settings
Presets\	your saved .gwpreset files

4. First Launch and Quick Start

1. Connect your GreaseWeazle(s) via USB and connect a floppy drive to the GreaseWeazle.
2. Start **GWCopyPro**. On startup the application automatically scans for GreaseWeazle devices ("Scanning for GreaseWeazle devices..." appears in the status bar) and queries the firmware version of each device it finds.
3. If your device appears in the **DEVICES** strip with a green pulsing LED — you are ready. If not, open  **Devices** and add it manually (see [chapter 6](#)).
4. Insert a disk into the drive.
5. Click ► **New Job**, choose *Read (disk → image)*, pick an image file name (e.g. `mydisk.scp`), optionally select a *Disk Format*, and press ► **Start Job**.
6. Watch the track grid fill up with green cells as `gw.exe` reads the disk, cylinder by cylinder.



5. The Main Window



The main window is divided into four areas, from top to bottom:

5.1 The toolbar

Button	Function
► New Job	Opens the New Job dialog without a pre-selected device.
⦿ Devices	Opens the Device Manager .
⚙ Settings	Opens the Settings dialog (gw.exe path, language).
✕ Clear Done	Removes all <i>Completed</i> , <i>Error</i> , and <i>Cancelled</i> job panels from the jobs list. Running jobs are never touched.

5.2 The DEVICES strip

A horizontal row of device cards, one per registered GreaseWeazle. Each card shows:

- the **device name** you assigned,
- the **COM port** (Port: COM3),
- the **firmware version** (FW: 1.6) as reported by gw.exe info,
- the **connection status** (● Connected / ● Disconnected),
- a **pulsing LED** — green and pulsing while connected, red when disconnected,
- a ► **New Job** button that opens the job dialog with this device pre-selected,
- a ⚡ **Blink** button that pulses the attached drive's LED three times, alternating drive 0 and drive 1, so you can tell which physical drive belongs to this card — handy when several GreaseWeazles are connected. It's disabled while the device is disconnected or while another identify sequence is already running; the status bar shows "*Blinking <device> (<port>)*..." while it runs.
- a ✕ button that removes the device from the list.

If no devices are registered, a hint text is shown instead: "*No devices added. Click ⦿ Devices to add a GreaseWeazle.*"



(Screenshot placeholder: a device card showing the new ⚡ Blink button next to ▶ New Job and ×.)

5.3 The ACTIVE JOBS area

A scrollable list of [job panels](#), one per job. The header shows a live counter, e.g. **3 total, 1 running**. Jobs remain listed after they finish (so you can inspect logs or restart them) until you press **× Clear Done**.

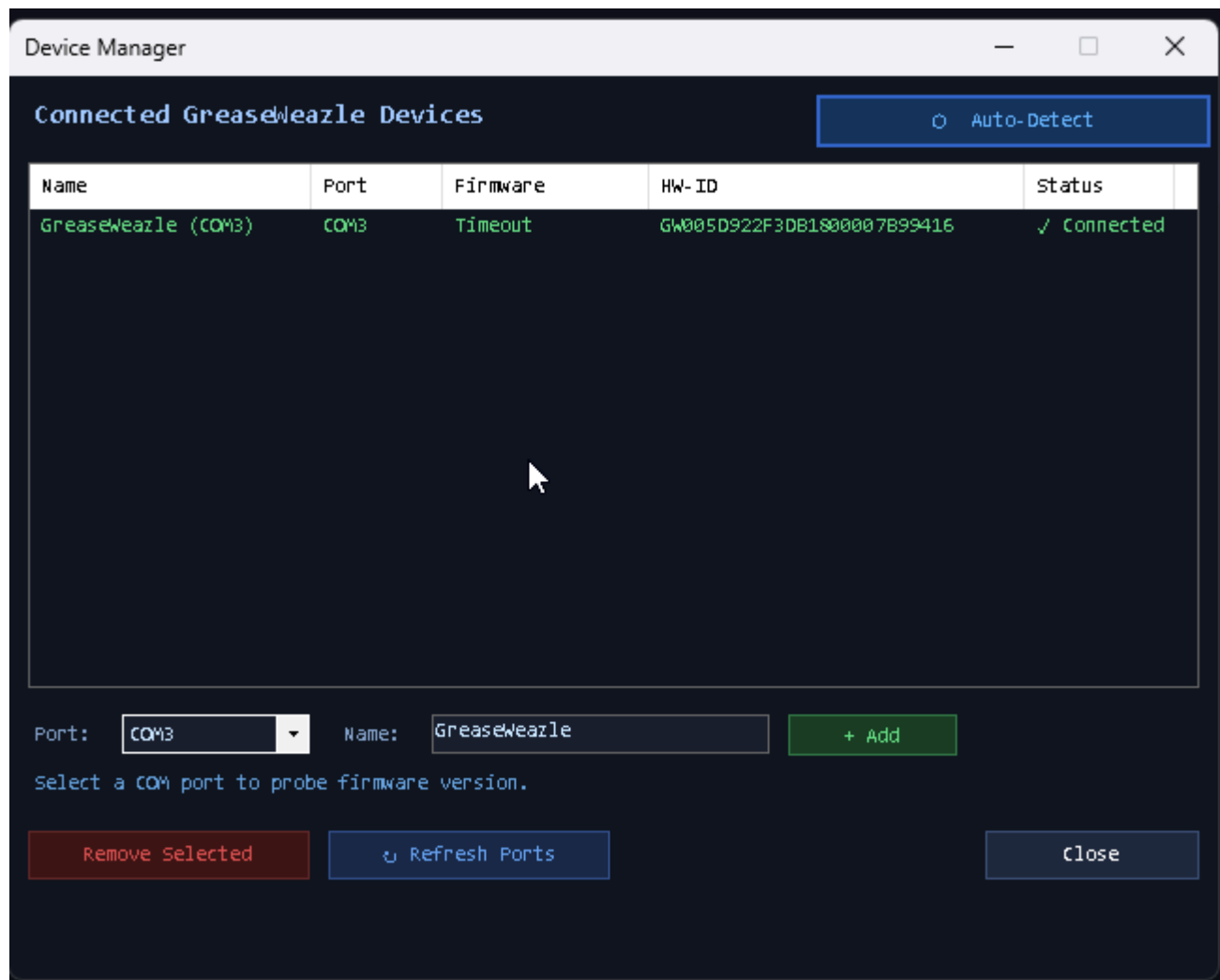
Multiple jobs may run **simultaneously** on different devices — there is no artificial limit. Every job runs in its own background thread.

5.4 The status bar

- **Left:** the currently configured path to `gw.exe` (`gw.exe: C:\gw\gw.exe`).
- **Right:** the most recent status message (job started/completed, device detection, errors). Messages automatically revert to "Ready" after 4 seconds.

6. The Device Manager

Open with **🔌 Devices** in the toolbar.



The Device Manager dialog lets you view, add, and remove GreaseWeazle devices **at runtime** — no application restart needed.

6.1 The device list

Columns: **Name**, **Port**, **Firmware**, **HW-ID** (the hardware ID reported by Windows), and **Status** (✓ Connected / Disconnected). Connected devices are shown in green, disconnected ones in red.

6.2 🔍 Auto-Detect

Click 🔍 **Auto-Detect** to scan the system (via Windows WMI) for connected GreaseWeazle hardware. For every newly found device, GWCOPYPro runs `gw.exe info --device COMx` in the background to query the firmware version, then adds the device to the list. Devices that are already registered are skipped ("All detected devices were already registered.").

6.3 Adding a device manually

If auto-detection does not find your device (e.g. unusual USB-serial adapter):

1. Select the **Port** from the drop-down (use ↻ **Refresh Ports** if your port is missing). As soon as you select a port, GWCOPYPro probes it and shows the firmware result (`COM3 → Firmware: 1.6`).
2. Enter a friendly **Name** (e.g. "GW #1 — 3.5 inch drive").
3. Click **+ Add**.

6.4 Removing a device

Select the device row and click **Remove Selected**. This only removes it from GWCopyPro's list — nothing happens to the hardware.

Click **Close** to return to the main window; the device strip refreshes automatically.

7. Creating a Job — the New Job Dialog

Open with ► **New Job** (toolbar or device card). This is the heart of GWCopyPro: five tabs cover every relevant **gw.exe** option, and a **live command preview** at the bottom of the dialog always shows the exact command line that will be executed, e.g.:

```
gw.exe read --device COM3 --format ibm.1440 --tracks=c=0-79:h=0-1 --retries 3
"C:\Images\disk1.img"
```

The preview updates instantly as you change any control — a great way to *learn* the **gw.exe** syntax while using the GUI.

At the bottom of the dialog you also find:

Button	Function
 Save Preset	Saves the complete current dialog state to a .gwpreset file (chapter 10).
 Load Preset	Loads a .gwpreset file and fills all controls from it.
► Start Job	Validates the input and starts the job.
Cancel	Closes the dialog without starting anything.

Validation: If you are *not* using Repetitive mode, an image file must be specified — otherwise the dialog shows *"Please specify an image file."* and stays open.

7.1 Tab "Main Settings"

New GreaseWeazle Job

Main Settings | Track Selection | Advanced | Post-Actions | Rep

Device: {Auto-select device}

Job Type: Read (disk → image)

Image File: ..

Disk Format:

— Common Options —

Revolutions (--revs): 1 (flux revs per track, default 1)

Density (--densel): {auto} Bitrate (--bitrate): 0 (0=auto)

— Read Options —

☐ --retries N: 3 ☐ --no-clobber ☐ --raw

☐ --reverse (flippy side B) ☐ --hard-sectors

— Write Options —

☐ --erase ☐ --verify ☐ --gen-tg43 (8" drives)

--precomp: ☐ --reverse ☐ --hard-sectors

gw.exe read --tracks=c=0-79:h=0-1

Save Preset Load Preset Start Job Cancel

Device

Selects which GreaseWeazle runs the job. (**Auto-select device**) lets `gw.exe` pick a device itself (fine if only one is connected). Selecting a specific device emits `--device COMx`.

Job Type

- **Read (disk → image)** — reads a physical floppy and saves it as an image file (`gw.exe read ...`).
- **Write (image → disk)** — writes an existing image file onto a physical floppy (`gw.exe write ...`).

The choice controls which option sections apply (Read Options vs. Write Options) and whether the file browser opens a *Save* dialog (read) or an *Open* dialog (write).

Image File

The full path of the disk image to create (read) or to write (write). The ... button opens a file dialog. Supported types in the browser: `*.scp`, `*.hfe`, `*.img`, `*.adf` (and `*.ipf` for writing). See the [Glossary](#) for what each type means.

Rule of thumb:

- For *archival* or *unknown/protected* disks: read to **.scp** (raw flux — preserves everything).
- For *known standard formats* you want to use in emulators: choose a *Disk Format* and read straight to **.img** (PC), **.adf** (Amiga), **.st/.img** (Atari ST) etc.

Disk Format

Corresponds to **--format**. Either type a format name directly into the text box, or pick one from the quick-select drop-down, which fills the text box for you. The list includes, among others:

Family	Formats
IBM PC	ibm.1440 , ibm.720 , ibm.1200 , ibm.360 , ibm.180 , ibm.320 , ibm.800 , ibm.2880
Amiga	amiga.amigados , amiga.amigados-hd
Atari ST	atarist.360 , atarist.400 , atarist.720 , atarist.800
Atari 8-bit	atari.90 , atari.130 , atari.180 , atari.360
Commodore	commodore.1541 , commodore.1571 , commodore.1581
Apple / Mac	apple2.525.ss.sd.35 , apple2.525.ss.sd.40 , mac.400 , mac.800
MSX	msx.1 , msx.2
NEC PC-98	pc98.2hd , pc98.2dd , pc98.2d
Acorn	acorn.adfs.s/m/l/d/e/f
DEC	dec.rx50 , dec.rx33
Ensoniq	ensoniq.mirage , ensoniq.esq1
Other	gem.1 , dragon.40 , coco.35 , zx.trdos.ds80

Leave the field **empty** to omit **--format** entirely (raw flux operation — typical when reading to **.scp**). The full list your gw.exe supports can be printed with **gw.exe read --help** or found in the GreaseWeazle wiki.

Common Options

Control	gw.exe flag	Meaning
Revolutions	--revs N	How many full disk revolutions of flux are captured per track. Default 1; higher values (2–5) give the decoder more chances to recover weak or damaged sectors and are needed for some copy-protection analysis.
Density	--dense1 hd/dd/ed	Overrides the density-select line to the drive. (auto) = let gw.exe decide. See <i>Density</i> in the glossary.
Bitrate	--bitrate N	Forces a specific data rate in kbit/s. 0 = auto-detect (recommended).

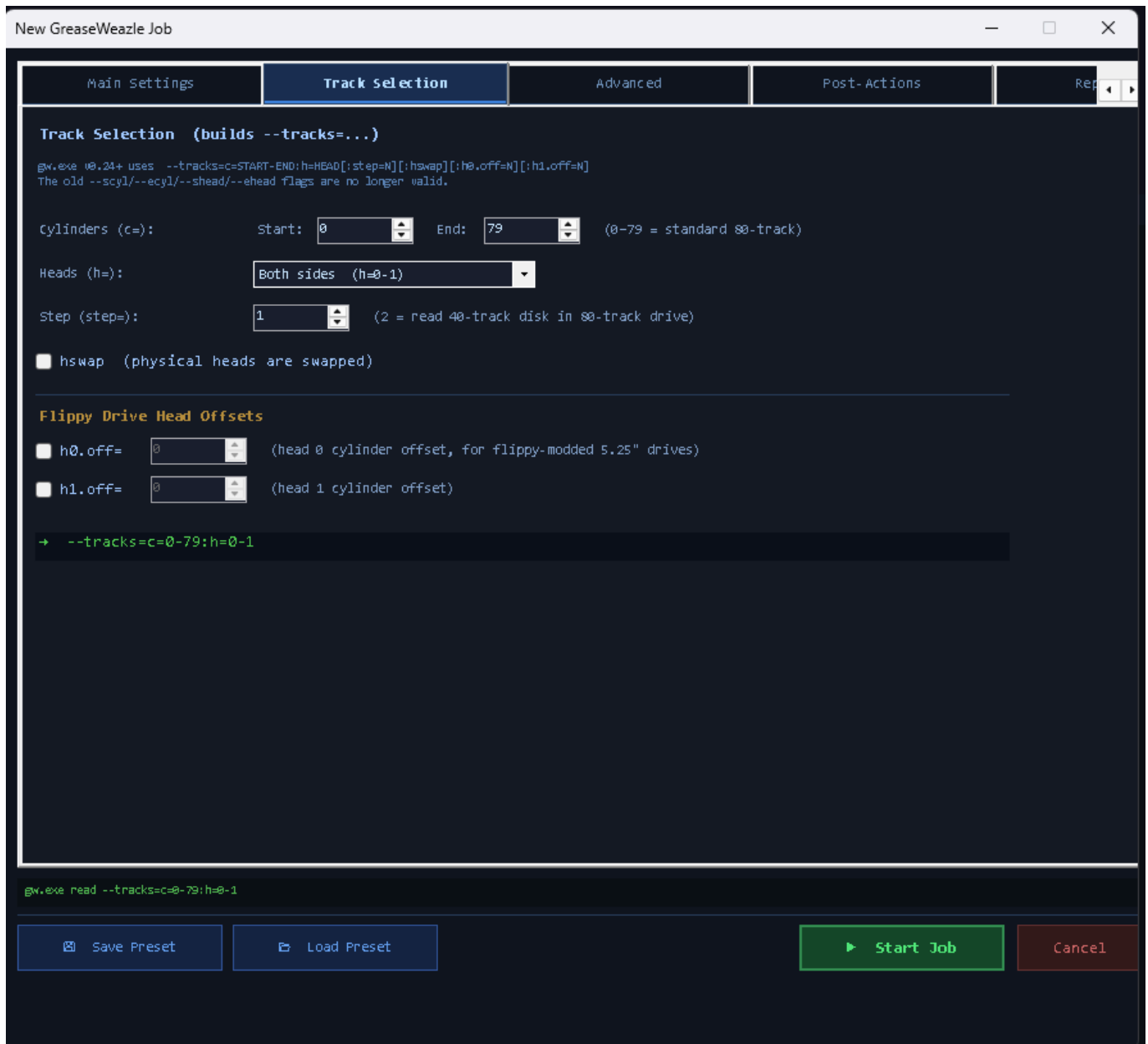
Read Options (only used for Read jobs)

Control	gw.exe flag	Meaning
--retries N	--retries N	Re-read a track up to N extra times when bad sectors are detected. Enable the checkbox and set the count (default 3).
--no-clobber	--no- clobber	Do not overwrite tracks that already exist in the output image — useful for <i>resuming</i> a partially completed read.
--raw	--raw	Capture raw flux without decoding it through the format codec.
--reverse (flippy side B)	--reverse	Reverses the track data — used when reading side B of a "flippy" disk in a flippy-modified drive.
--hard-sectors	--hard- sectors	Enables support for hard-sectored disks (multiple index holes).

Write Options (only used for Write jobs)

Control	gw.exe flag	Meaning
--erase	--erase	Erases each track before writing — recommended when the target disk previously held a different (especially higher-density) format.
--verify	--verify	Reads each track back after writing and compares — strongly recommended for important disks.
--gen-tg43 (8" drives)	--gen- tg43	Generates the /TG43 ("Track Greater than 43") signal needed by some 8" drives to reduce write current on inner tracks.
--precomp	--precomp N	Write precompensation in microseconds — shifts flux transitions slightly to counteract magnetic "bit shift" on inner tracks. Leave empty for default.
--reverse	--reverse	As above, for writing flippy side B.
--hard-sectors	--hard- sectors	As above, for hard-sectored media.

7.2 Tab "Track Selection"



This tab builds the compound `--tracks=` specifier introduced in `gw.exe` v0.24:

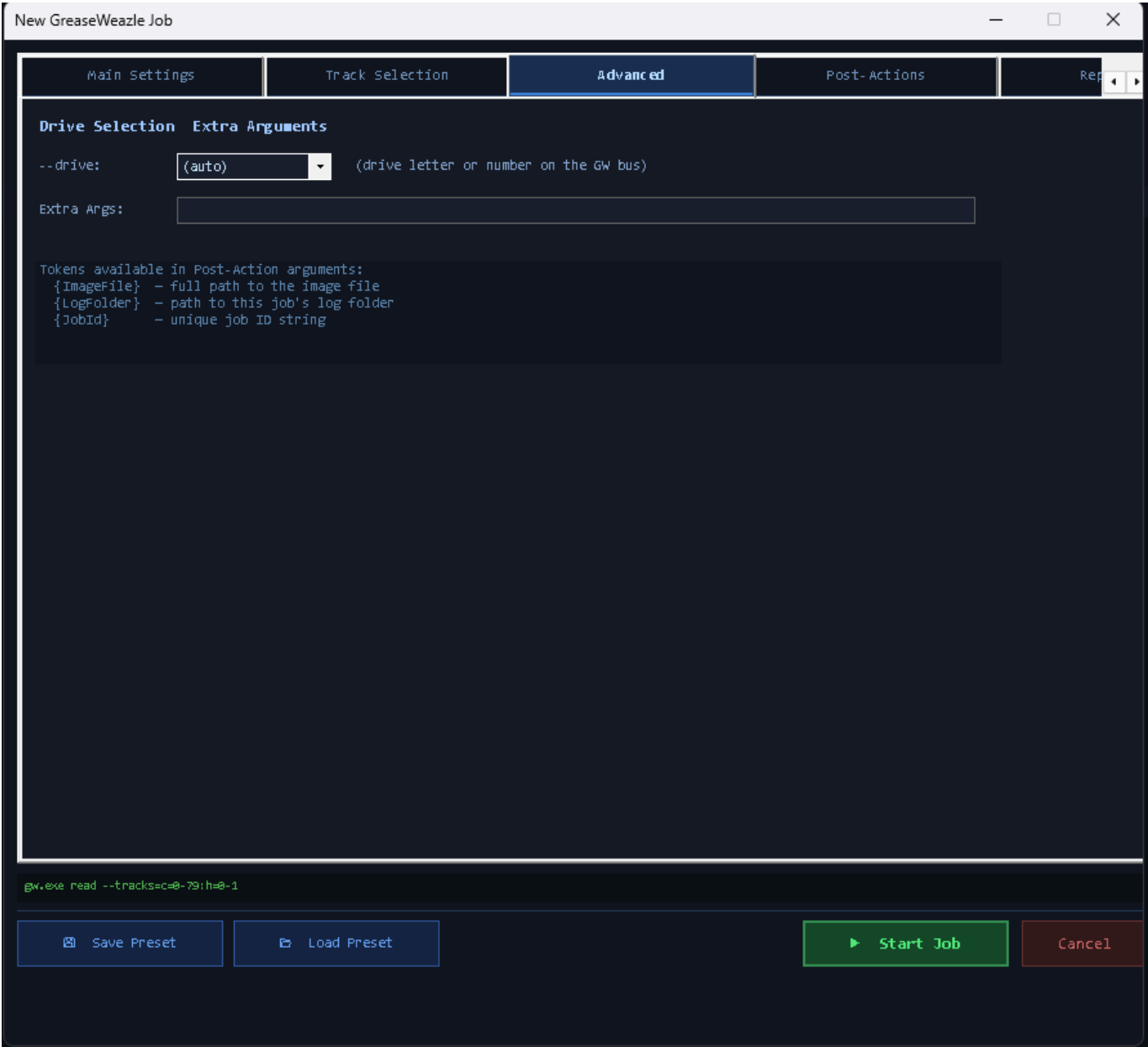
```
--tracks=c=START-END:h=HEAD[:step=N][:hswap][:h0.off=N][:h1.off=N]
```

The old flags `--scyl`, `--ecyl`, `--shead`, `--ehead`, `--single-sided` were **removed** in v0.24 and are never generated by GWCOPYPro.

A green preview line at the bottom of the tab (e.g. `→ c=0-79:h=0-1`) always shows the resulting specifier. When everything is at its default, GWCOPYPro omits `--tracks=` entirely and `gw.exe` processes the full disk on both sides.

Control	Spec component	Meaning
Cylinders Start / End	<code>c=0-79</code>	First and last cylinder to process (inclusive). 0–79 is a standard 80-track disk; 0–39 for 40-track (5.25" DD) disks; up to 83 for "overdumps".
Heads	<code>h=0-1</code> , <code>h=0</code> , <code>h=1</code>	<i>Both sides, Head 0 only</i> (bottom side), or <i>Head 1 only</i> (top side). Single-sided formats need only head 0.
Step	<code>step=2</code>	Physical head steps per logical cylinder. step=2 is the classic trick for reading a 40-track disk in an 80-track drive — the drive steps twice per data track.
hswap	<code>hswap</code>	Swaps the meaning of head 0 and head 1 — for drives with physically swapped head wiring.
h0.off= / h1.off=	<code>h0.off=N</code> / <code>h1.off=N</code>	Cylinder offset per head (–9...+9), used with flippy-modified 5.25" drives where one head is physically offset by a few cylinders. Enable the checkbox to activate.

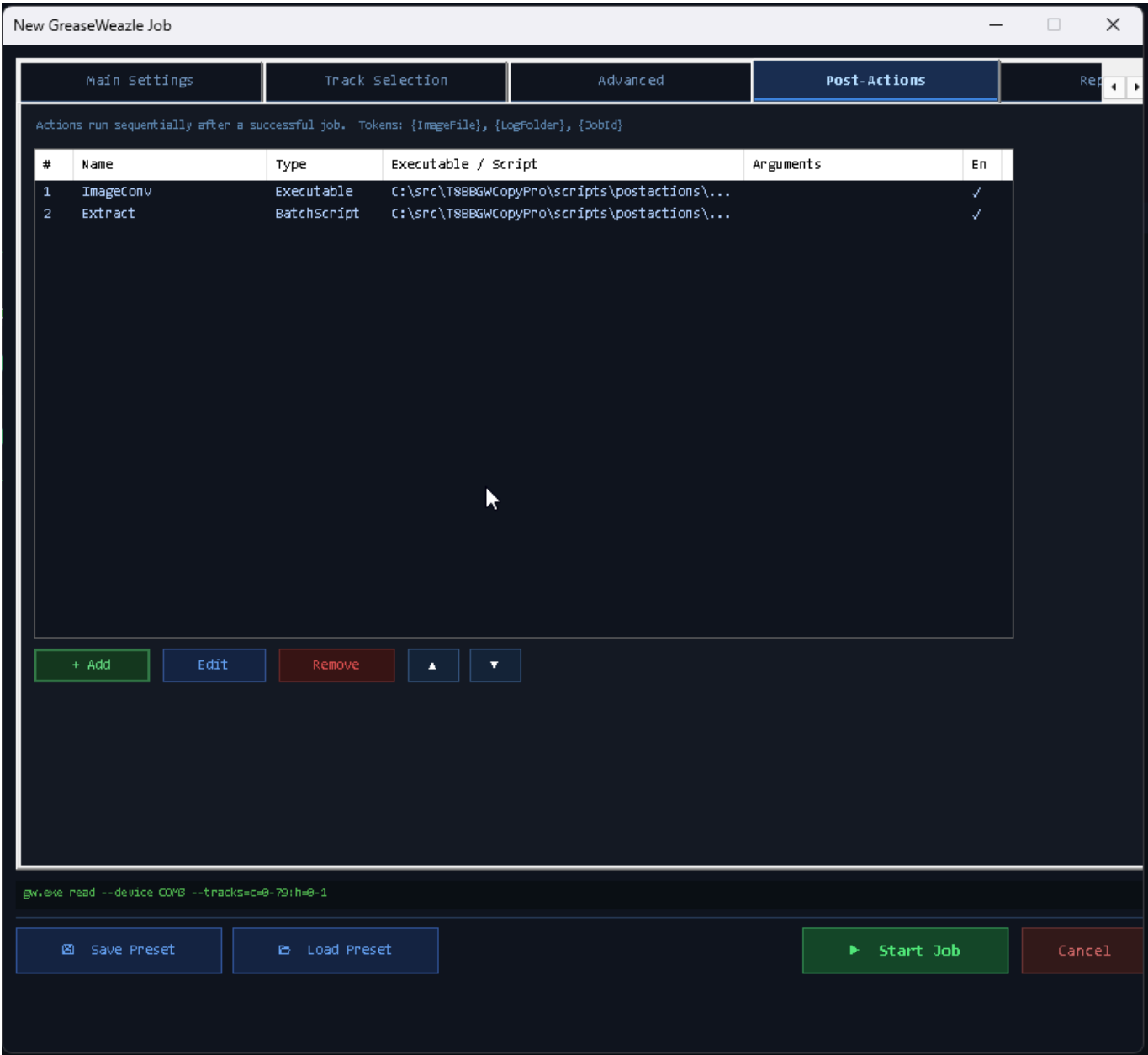
7.3 Tab "Advanced"



Control	Meaning
--drive	Which drive on the GreaseWeazle's floppy bus to use: a/b (IBM-style twisted cable) or 0-3 (Shugart-style straight cable). (auto) omits the flag and uses gw.exe's default (drive 0 / A).
Extra Args	A free-text field appended verbatim to the end of the command line, before the image file. Use it for any gw.exe option that has no dedicated control, e.g. --fake-index=200ms , --dd 0 , --seek-retries 5 , or format-specific options like --adjust-speed .

The tab also lists the tokens available in post-action arguments (see next section).

7.4 Tab "Post-Actions"



Post-actions are programs or scripts that GWCOPYPro runs **automatically and sequentially after each successful job** (and, in repetitive mode, after **each successful disk**). Failed or cancelled jobs do **not** trigger post-actions.

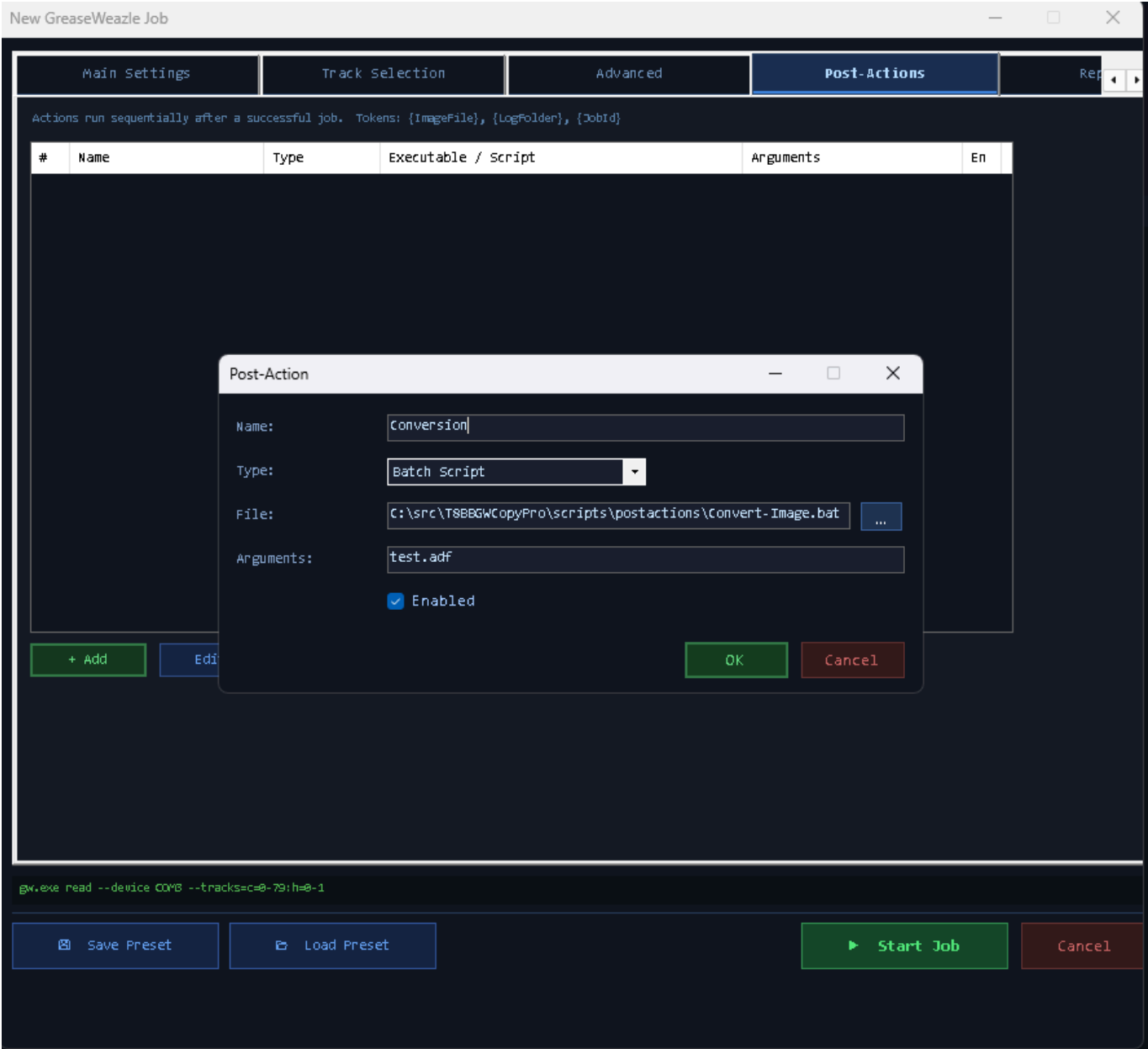
Typical uses: checksum generation, validation, zipping, copying to a NAS, converting flux images to sector images, extracting archives — ready-made scripts for all of these are in [chapter 14](#).

The action list

Columns: # (execution order), **Name**, **Type**, **Executable / Script**, **Arguments**, **En** (enabled ✓ / disabled —).

Button	Function
+ Add	Opens the Post-Action editor for a new action.
Edit	Edits the selected action.
Remove	Deletes the selected action.
▲ / ▼	Moves the selected action up/down in the execution order.

The Post-Action editor



Field	Meaning
Name	Display name in the list (e.g. "Validate image").
Type	How the action is launched — see table below.
File	Path to the <code>.exe</code> , <code>.bat</code> , or <code>.ps1</code> file.
Arguments	Argument string; may contain tokens (see below).
Enabled	Unchecking skips the action without deleting it.
Type	How it is executed
Executable	<code>yourfile.exe <arguments></code> — called directly.
Batch Script	<code>cmd.exe /c "yourscript.bat" <arguments></code>

Type	How it is executed
PowerShell Script	<code>powershell.exe -NoProfile -ExecutionPolicy Bypass -File "yourscript.ps1" <arguments></code>

Tokens

The following tokens in the *Arguments* field are replaced at runtime:

Token	Expands to
<code>{ImageFile}</code>	Full path of the disk image the job just produced/used.
<code>{LogFolder}</code>	Full path of the job's log folder.
<code>{JobId}</code>	The unique 8-character job ID.
<code>{DiskIndex}</code>	The current disk number (repetitive mode; <code>1</code> otherwise).

Always put quotes around path tokens — e.g. `"{ImageFile}"` — otherwise paths that contain spaces will break the argument parsing of your script.

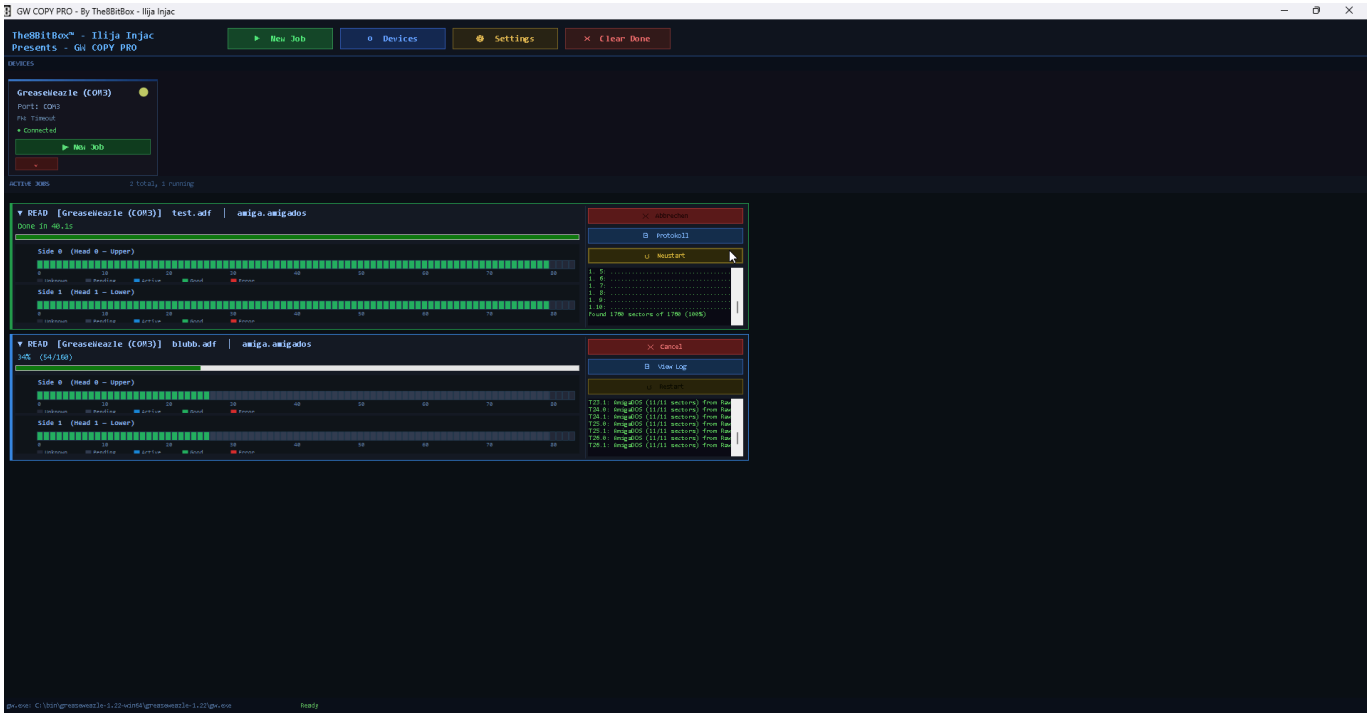
All output (stdout and stderr) of every post-action is appended to the job's `gw_output.log`, together with the action's exit code — so you can always verify what happened.

7.5 Tab "Repetitive"

Described in detail in [chapter 9](#). This tab also contains the **Preset Name** field used when saving presets, and the **Use device group** checkbox for running the job across several devices in parallel — see [9.3 Group Repetitive Jobs — Parallel Batch Imaging](#).

8. The Job Panel and the Disk Visualiser

Every started job gets its own panel in the ACTIVE JOBS area.



8.1 Panel contents

- **Title line** — job type, image file name, device.
- **Status line** — e.g. **45% (72/160)**, or in repetitive mode **Disk #3 45% (72/160)**; after completion **Done in 92.4s**, on failure **Error: gw.exe exited with code 1**.
- **Progress bar** — overall percentage of tracks completed.
- **Two track grids** — *Side 0 (Head 0 — Upper)* and *Side 1 (Head 1 — Lower)*, each an 84-cell bar, one cell per cylinder.
- **Live log pane** — the most recent **gw.exe** output lines, scrolling in real time.
- **Buttons:**

Button	Function
✕ Cancel	Cancels this job: the gw.exe process (and its children) is terminated and the job is marked <i>Cancelled</i> .
📄 View Log	Opens the job's log folder in Windows Explorer (or the log file in Notepad if only the file exists).
↺ Restart	Available once the job has finished — re-opens the New Job dialog pre-filled with this job's exact configuration (from its preset snapshot).

8.2 Cell colours

Colour	Status	Meaning
Dark grey	Unknown	Not part of the selected track range / not started.
Mid grey	Pending	Queued, not yet processed.
Blue	Reading/Writing	Currently being processed.
Green	Good	Track finished successfully.

Colour	Status	Meaning
Red	Error	gw.exe reported an error on this track (bad sectors, etc.).

The cells are driven by parsing `gw.exe` output lines such as `T12.0: ok` or `Cyl 12, Head 0: reading`. For `gw.exe` versions with different output wording, a fallback parser tracks `n/m` progress fractions so the progress bar keeps moving even if individual cells cannot be attributed.

A few **red cells** do not necessarily mean a failed job: if `gw.exe` exits with code 0, the job is *Completed* — red cells then mark tracks that had read errors despite retries. Check the log for details and consider re-reading with more `--retries` or `--revs`.

9. Repetitive Mode — Imaging Whole Boxes of Disks

Repetitive mode is made for bulk digitisation sessions: you configure a job **once**, and GWCOPYPro loops it disk after disk, numbering the files automatically.

9.1 Configuration (tab "Repetitive")

New GreaseWeazle Job

Track Selection Advanced Post-Actions **Repetitive**

☐ Repetitive mode (image sequence)

Output Folder: ...

Tokens: {n} {n:d3} {dt} Example: Disk_{n:d3}_{dt}.scp

Start Index:

DateTime format: (C# DateTime format, e.g. yyyyMMdd_HH:mm:ss)

Preview:

The file pattern overrides the image file for each disk.
After each disk a dialog will ask to insert the next one.

Preset Name:

gw.exe read --device COWD --drive 1 --tracks=c=0-79:h=0-1

1. Check **Repetitive mode (image sequence)**.
2. Choose an **Output Folder** (leave empty to use the image file's folder; if neither is set, the Desktop is used).
3. Enter a **File Pattern** using tokens:

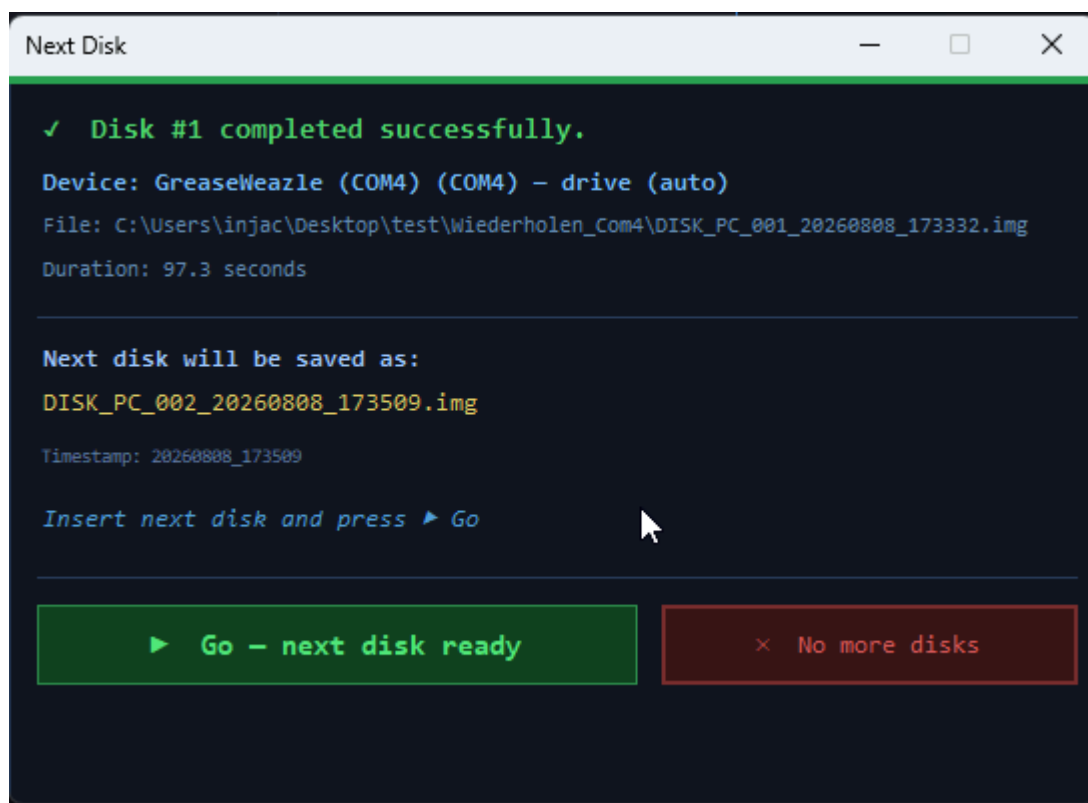
Token	Meaning	Example (Disk_{n:D3}_{dt}.scp , disk 7)
{n}	Disk counter as plain number	Disk_7_....scp
{n:D3}	Counter with .NET format (D3 = 3 digits, zero-padded)	Disk_007_....scp
{dt}	Timestamp using the <i>DateTime</i> format field	Disk_007_20260728_143022.scp

4. Set the **Start Index** (first counter value, default 1) and, if desired, adjust the **DateTime format** (a C# *DateTime* format string, default `yyyyMMdd_HH:mm:ss`).
5. The **Preview** line shows exactly what the first file will be called.

The file pattern **overrides** the *Image File* field for each disk. If the pattern contains no token at all, repetitive mode is not engaged and a normal single job runs.

9.2 The loop

1. The job runs for disk #1 exactly like a normal job (including post-actions).
2. When the disk completes, the **Next Disk dialog** appears:



- a **Device: <name> - drive <address>** line identifying which GreaseWeazle and drive produced this disk (and will produce the next one); the drive shows **(auto)** if none was explicitly selected,
- ✓ **Disk #3 completed successfully.** — with the written file name and duration,

- the file name the **next** disk will get,
 - a pulsing prompt: *"Insert next disk and press ► Go"*.
3. Swap the disk, press ► **Go — next disk ready** — the counter increments and the next disk is imaged. Press ✕ **No more disks** to end the session; the job is then marked *Completed* with a summary (**Done — 12 disk(s) in ...**).

The track grid resets for every disk, and each disk gets its **own log folder** (*..._disk1, ..._disk2, ...*). Post-actions run after **every** disk, with `{DiskIndex}` carrying the current number — ideal for per-disk validation or archiving.

This device/drive line only appears for **single-device** repetitive jobs. Group jobs (below) never show the Next Disk dialog — the Batch Insert dialog carries the per-device info instead.

9.3 Group Repetitive Jobs — Parallel Batch Imaging

For digitising a large collection faster, repetitive mode can target a **device group** instead of a single device: several GreaseWeazle devices image one batch of disks at the same time, guided by an LED blink-and-verify sequence so you always insert the right disk into the right drive.

Setting up a group

On the **Repetitive** tab, below the normal repetitive controls:

1. Check **"Use device group (parallel batch imaging)"** — enabled only while Repetitive mode is also checked. The single Device/Drive selectors elsewhere in the dialog are disabled while a group is active; the group member list replaces them.
2. For each drive you want in the group, pick a **Device** and a **Drive** (*0 / 1 / a / b*) and click **+ Add**; the pair is appended to the member list, in the order the drives will blink. Select a row and click **– Remove** to take it out again.
3. A group needs **at least two members**, and the same physical GreaseWeazle device may not appear twice — parallel imaging needs one device per drive, since *gw.exe* holds a COM port exclusively.

New GreaseWeazle Job

Track Selection Advanced Post-Actions **Repetitive**

☒ **Repetitive mode (image sequence)**

Output Folder:

Tokens: {n} {n:D3} {dt} Example: Disk_{n:D3}_{dt}.scp

Start Index:

DateTime format: (C# DateTime format, e.g. yyyyMMdd_HH:mm:ss)

Preview:

The file pattern overrides the image file for each disk.
After each disk a dialog will ask to insert the next one.

Preset Name:

☒ **Use device group (parallel batch imaging)**

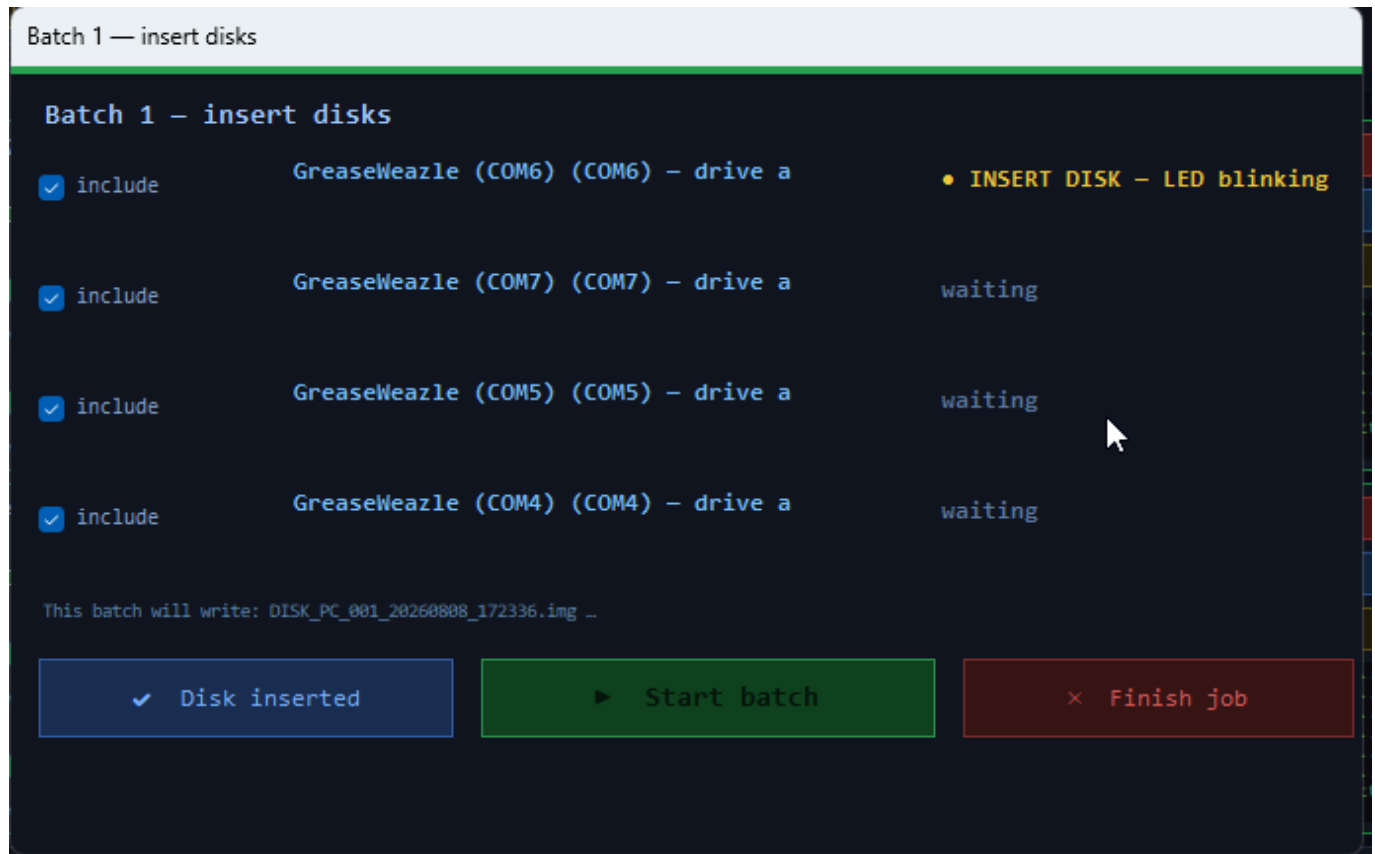
Device	Drive
GreaseWeazle (COM6) (COM6)	a
GreaseWeazle (COM7) (COM7)	a
GreaseWeazle (COM5) (COM5)	a
GreaseWeazle (COM4) (COM4)	-

gw.exe read --device COM7 --format ibm.1440 --tracks=c=0-79:h=0-1

Group configuration round-trips through job presets exactly like any other setting — including when a saved preset references a device that is no longer connected.

The Batch Insert dialog

Starting a group job — and again after every completed batch — opens the **Batch Insert dialog**, which replaces the ordinary Next Disk dialog for group jobs only:



- The header shows the batch number and a preview of the file names this batch will produce.
- One row per group member — device name, COM port and drive, an **include** checkbox, and a status:

Status	Meaning
waiting	Not yet this row's turn to blink.
• INSERT DISK — LED blinking	This drive's LED is pulsing now — insert a disk into it.
✓ disk detected	You confirmed a disk is inserted in this drive.
— excluded	This drive is skipped for the current batch.

A row also shows the previous batch's outcome once available (**last:** ✓ <filename> or **last:** X <error>).

- **Only one drive blinks at a time**, in member-list order, skipping excluded rows. Click **✓ Disk inserted** once you've inserted a disk into the currently blinking drive to advance immediately to the next included, unverified row. GWCopyPro does not probe the drive for a disk before advancing — gw.exe has no reliable, dedicated way to detect disk presence, so your confirmation is trusted as-is. If a drive turns out to be empty or otherwise fails, that member's imaging simply reports an error without blocking the rest of the batch (see below).
- Unchecking **include** on the currently blinking row skips it immediately; re-checking it later appends it back to the end of the blink queue — exclusion is per batch and fully reversible.
- **▶ Start batch** is enabled only once every included drive reads **disk detected**. All included drives then image their disk **in parallel**, each getting its own job panel and live track visualiser in the main window, exactly like a normal job.
- If one drive's disk fails, the others are unaffected and keep imaging; the failure is reported on that drive's row in the next batch, and its disk number is never reused for a later disk.

- ✕ **Finish job** ends the group session at any point (also triggered by closing the dialog); the job is then marked *Completed* with a summary of all disks written.

Post-actions run per finished disk exactly as in a normal job, with `{DiskIndex}` carrying that disk's number.

Tip: you don't have to fill every drive on every batch — exclude a drive that's temporarily empty (e.g. you haven't sorted its next disk yet) and re-include it on a later batch.

Cancelling a group job stops every member's `gw.exe` process, just like cancelling a normal job.

10. Job Presets

A preset stores **everything** in the New Job dialog: device, job type, format, all flags, track selection, post-actions, and the complete repetitive configuration.

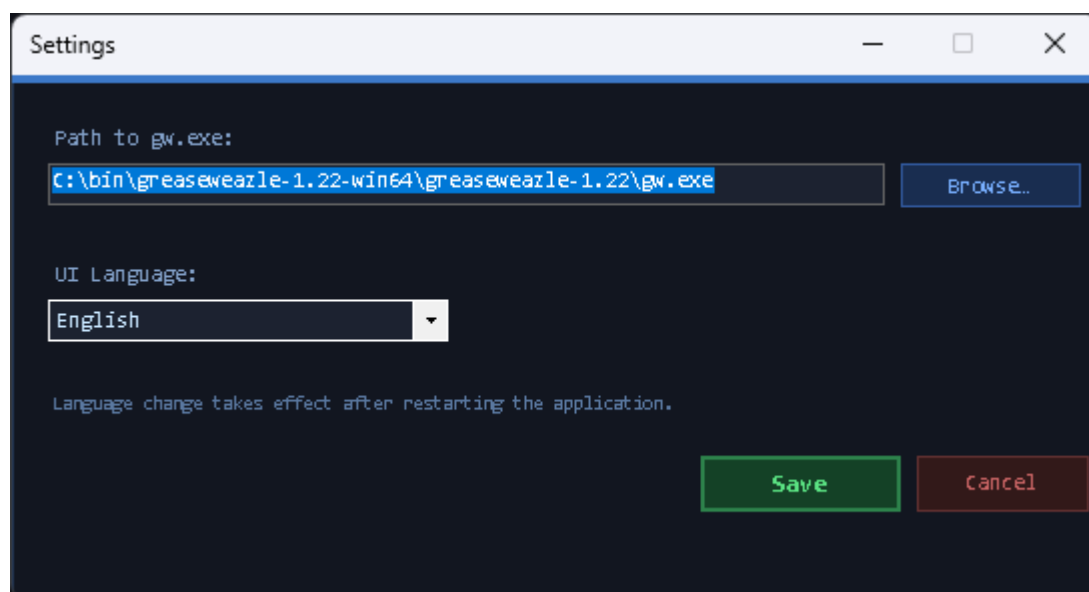
-  **Save Preset** (New Job dialog) — writes a JSON file with extension `.gwpreset`, by default into `%APPDATA%\GreaseWeazleManager\Presets\`. The file name is derived from the **Preset Name** field on the *Repetitive* tab.
-  **Load Preset** — loads a `.gwpreset` file and fills every control from it.
-  **Restart** (job panel) — every started job keeps an internal snapshot of its configuration; Restart re-opens the dialog pre-filled with it, even if you never saved a preset file.

Because presets are plain JSON, you can inspect, back up, or share them freely.

Example workflow: create presets "Amiga DD → ADF", "PC 1.44MB → IMG", and "Unknown disk → SCP raw archive (3 revs)" once — from then on any imaging session starts with two clicks.

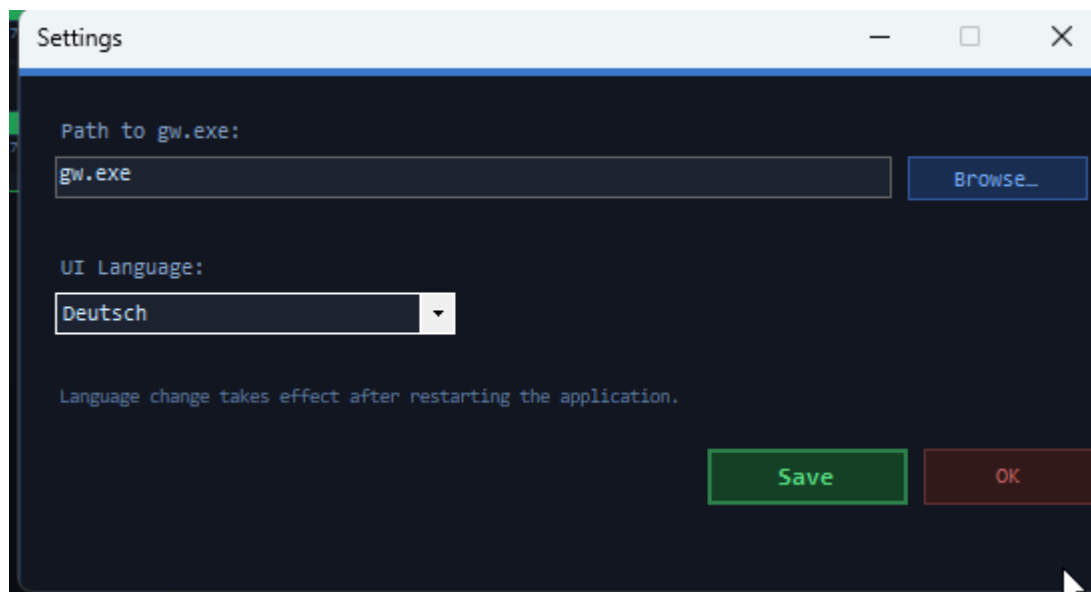
11. Settings

Open with ⚙ **Settings**.



Setting	Meaning
Path to gw.exe	Full path to your gw.exe . Default is plain gw.exe , which works when the gw tools folder is on your PATH . Use Browse... to pick the file.
UI Language	English or Deutsch. The change is applied on Save ; some elements only refresh fully after restarting the application.

After you click **Save**, the **Cancel** button relabels itself **OK** — a quick visual confirmation that your changes were written to disk. It still just closes the dialog (nothing is undone), so there is no separate "apply" step to remember.



Settings are stored in `%APPDATA%\GreaseWeazleManager\settings.json` and persist across sessions.

12. Logging

Every job writes a complete log:

```
<application folder>\Logs\
  Job_Read_a1b2c3d4_20260728_143022\      ← one folder per job
    gw_output.log
  Job_Read_e5f6a7b8_20260728_150001_disk1\ ← repetitive mode: one folder per
disk
  Job_Read_e5f6a7b8_20260728_150001_disk2\
```

gw_output.log contains:

1. A header with job type, device, disk number, the **full gw.exe command line**, and the start time.
2. Every stdout line of **gw.exe**, live; stderr lines prefixed **[ERR]**.
3. The completion marker (**[COMPLETED]**, **[CANCELLED]**, **[ERROR]** Exit code: **N**, or **[EXCEPTION]** ...).
4. A **=== Post-Actions ===** section with each action's command, its output, and its exit code (**[ACTION]** Exit: **0**).

The  **View Log** button on the job panel takes you straight to the folder.

Tip: the log always contains the exact command line — copy it into a terminal to reproduce or fine-tune a run manually.

13. Audio and Visual Feedback

GWCopyPro signals important events audibly (useful when you are sorting disks and not watching the screen):

Event	Sound	Visual
Job started	Two ascending beeps	Status bar message
Job / disk completed	Three ascending beeps	Green status message, green track cells
Job error	Three descending beeps	Red status message + the main window background flashes red four times
Track error	—	Red cell in the visualiser

14. Post-Action Script Cookbook

This chapter contains **ready-to-use scripts** for the Post-Actions tab. Copies of all scripts are shipped in the repository folder `scripts/postactions/` — copy them anywhere you like (e.g. next to `GWCopyPro.exe` in a `scripts\` folder) and reference them from the Post-Action editor.

General configuration pattern (Post-Action editor):

Field	Value
Type	<i>PowerShell Script</i> (for <code>.ps1</code>) or <i>Batch Script</i> (for <code>.bat</code>)
File	full path to the script
Arguments	as given per recipe below — keep the quotes around tokens!

All scripts write their progress to stdout, which lands in the job's `gw_output.log`.

14.1 Extracting archives with the bundled lsar.exe / unar.exe

GWCopyPro ships the command-line tools of *The Unarchiver* in its `tools\` folder:

- **lsar.exe** — *lists* the contents of an archive (zip, rar, 7z, lha/lzh, adz, ...).
- **unar.exe** — *extracts* archives of nearly any type.

This is handy when your disk images (or the software you want to write to disk) arrive inside archives: `.zip`, `.rar`, `.lha` (very common in the Amiga world), `.7z`, and even `.adz` (gzipped ADF).

Basic manual usage:

```
lsar.exe "C:\Downloads\games.rar"           List contents
unar.exe -force-overwrite -o "C:\Extracted" "games.rar"  Extract everything to
C:\Extracted
```

Script: **Extract-Archive.ps1**

Lists an archive with **lsar.exe**, extracts it with **unar.exe**, and reports every disk image (**.adf**, **.scp**, **.img**, **.st**, **.hfe**, **.ipf**, ...) it finds:

```
<# Extract-Archive.ps1
Lists an archive with lsar.exe and extracts it with unar.exe.
Reports all disk images found after extraction.

Post-Action setup:
  Type:      PowerShell Script
  File:      C:\...\scripts\postactions\Extract-Archive.ps1
  Arguments: -Archive "{ImageFile}" -Destination "D:\Extracted"

#>
param(
  [Parameter(Mandatory = $true)][string]$Archive,
  [string]$Destination = "",
  [string]$ToolsDir    = ""
)

$ErrorActionPreference = "Stop"

# Locate lsar/unar: explicit -ToolsDir, then tools\ next to this script's
# grandparent (repo layout), then tools\ next to GWCopyPro.exe, then PATH.
function Find-Tool([string]$name) {
  $candidates = @()
  if ($ToolsDir) { $candidates += (Join-Path $ToolsDir $name) }
  $candidates += (Join-Path $PSScriptRoot "..\..\tools\$name")
  $candidates += (Join-Path (Split-Path $PSScriptRoot -Parent) "tools\$name")
  $candidates += (Join-Path (Get-Location) "tools\$name")
  foreach ($c in $candidates) { if (Test-Path $c) { return (Resolve-Path
$c).Path } }
  $cmd = Get-Command $name -ErrorAction SilentlyContinue
  if ($cmd) { return $cmd.Source }
  throw "$name not found. Pass -ToolsDir <folder containing lsar.exe/unar.exe>."
}

$lsar = Find-Tool "lsar.exe"
$unar = Find-Tool "unar.exe"

if (-not (Test-Path $Archive)) { throw "Archive not found: $Archive" }

if (-not $Destination) {
  $Destination = Join-Path (Split-Path $Archive -Parent) `
    ([IO.Path]::GetFileNameWithoutExtension($Archive))
}
```

```
New-Item -ItemType Directory -Force -Path $Destination | Out-Null

Write-Output "=== Archive contents ($([IO.Path]::GetFileName($Archive))) ==="
& $lsar $Archive

Write-Output "=== Extracting to $Destination ==="
& $unar -force-overwrite -output-directory $Destination $Archive
if ($LASTEXITCODE -ne 0) { throw "unar.exe failed with exit code $LASTEXITCODE" }

$imageExt = ".adf", ".adz", ".scp", ".img", ".ima", ".st", ".hfe", ".ipf", ".d64",
".dsk"
$images = Get-ChildItem $Destination -Recurse -File |
    Where-Object { $imageExt -contains $_.Extension.ToLower() }

Write-Output "=== Disk images found: $($images.Count) ==="
$images | ForEach-Object { Write-Output "  $_.FullName  ($($_.Length) bytes)" }
exit 0
```

Typical uses:

- **Preparing write jobs:** you keep `.lha/.zip` archives of Amiga software; run the script manually (or as a post-action of a "dummy" job) to unpack the `.adf` files, then write them with a Write job.
- **Post-processing downloads dropped into the output folder.**

Note on `unar.exe`: the user-requested `unar.exe` is *not* bundled — it is the free command-line extractor from the WinRAR makers ([rarlab.com](https://www.rarlab.com)). `unar.exe` already extracts RAR archives, so you normally don't need it. If you prefer unrar anyway, see the next recipe.

Script: `Extract-Rar.bat` (using `unar.exe`)

```
@echo off
REM Extract-Rar.bat – extracts a RAR archive with unrar.exe
REM
REM Requires unrar.exe (https://www.rarlab.com/rar_add.htm) – either on PATH
REM or adjust the UNRAR variable below.
REM
REM Post-Action setup:
REM   Type:      Batch Script
REM   File:      C:\...\scripts\postactions\Extract-Rar.bat
REM   Arguments: "{ImageFile}" "D:\Extracted"
REM
set "UNRAR=unar.exe"

if "%~1"=="" (
    echo Usage: Extract-Rar.bat archive.rar [destination]
    exit /b 2
)

set "DEST=%~2"
if "%DEST%"=="" set "DEST=%~dp1%~n1"
if not exist "%DEST%" mkdir "%DEST%"
```

```

echo === Listing %~nx1 ===
"%UNRAR%" 1 "%~1"

echo === Extracting to %DEST% ===
"%UNRAR%" x -y -o+ "%~1" "%DEST%"
if errorlevel 1 (
    echo [ERROR] unrar failed with exit code %errorlevel%
    exit /b 1
)
echo Done.
exit /b 0

```

14.2 Zipping images and output folders

Script: Zip-Image.ps1 — compress the image the job just produced

```

<# Zip-Image.ps1
Compresses the finished disk image into a .zip placed next to it.

Post-Action setup:
Type:      PowerShell Script
File:      C:\...\scripts\postactions\Zip-Image.ps1
Arguments: -ImageFile "{ImageFile}"
Optional:  add -DeleteOriginal to remove the uncompressed image
afterwards.
#>
param(
    [Parameter(Mandatory = $true)][string]$ImageFile,
    [switch]$DeleteOriginal
)

$ErrorActionPreference = "Stop"
if (-not (Test-Path $ImageFile)) { throw "Image not found: $ImageFile" }

$zip = [IO.Path]::ChangeExtension($ImageFile, ".zip")
Compress-Archive -Path $ImageFile -DestinationPath $zip -CompressionLevel Optimal
-Force

$src = (Get-Item $ImageFile).Length
$dst = (Get-Item $zip).Length
Write-Output ("Zipped {0} -> {1} ({2:N0} -> {3:N0} bytes, {4:P0} of original)" -f
,
    [IO.Path]::GetFileName($ImageFile), [IO.Path]::GetFileName($zip), $src, $dst,
($dst / $src))

if ($DeleteOriginal) {
    Remove-Item $ImageFile
    Write-Output "Original image deleted."
}

```

```
}
exit 0
```

Script: **Zip-OutputFolder.ps1** — archive the whole output directory

Perfect as the *last* action of a repetitive session, or run manually after a batch:

```
<# Zip-OutputFolder.ps1
Compresses ALL disk images in a folder into one timestamped zip archive.

Post-Action setup:
  Type:      PowerShell Script
  File:      C:\...\scripts\postactions\Zip-OutputFolder.ps1
  Arguments: -Folder "D:\FloppyImages"
             (or:      -Folder "{ImageFile}" to use the folder the image lives in)
#>
param(
  [Parameter(Mandatory = $true)][string]$Folder,
  [string]$ZipPath = ""
)

$ErrorActionPreference = "Stop"

# Accept either a folder or a file (then its parent folder is used)
if (Test-Path $Folder -PathType Leaf) { $Folder = Split-Path $Folder -Parent }
if (-not (Test-Path $Folder)) { throw "Folder not found: $Folder" }

if (-not $ZipPath) {
  $stamp = Get-Date -Format "yyyyMMdd_HH:mm:ss"
  $ZipPath = Join-Path $Folder ("Images_{0}.zip" -f $stamp)
}

$imageExt = ".adf", ".adz", ".scp", ".img", ".ima", ".st", ".hfe", ".ipf", ".d64",
".dsk"
$files = Get-ChildItem $Folder -File |
  Where-Object { $imageExt -contains $_.Extension.ToLower() }

if ($files.Count -eq 0) { Write-Output "No disk images found in $Folder - nothing
to do."; exit 0 }

Compress-Archive -Path $files.FullName -DestinationPath $ZipPath -CompressionLevel
Optimal -Force
Write-Output ("Archived {0} image(s) into {1}" -f $files.Count, $ZipPath)
exit 0
```

14.3 Validating images

Script: **Validate-Image.ps1** — sanity check + SHA-256 checksum

Checks that the image exists, is not empty, has a plausible size for its type, and writes a `.sha256` checksum sidecar file for long-term integrity verification:

```
<# Validate-Image.ps1
Validates a freshly created disk image:
  1. File exists and is not zero bytes.
  2. File size matches the expected size for known image types (warning only).
  3. Writes a SHA-256 checksum to "<image>.sha256".
Exit code 0 = OK, 1 = validation failed (visible in gw_output.log).

Post-Action setup:
  Type:      PowerShell Script
  File:      C:\...\scripts\postactions\Validate-Image.ps1
  Arguments: -ImageFile "{ImageFile}"

#>
param(
    [Parameter(Mandatory = $true)][string]$ImageFile
)

$ErrorActionPreference = "Stop"

if (-not (Test-Path $ImageFile)) {
    Write-Output "[FAIL] Image file does not exist: $ImageFile"
    exit 1
}

$item = Get-Item $ImageFile
if ($item.Length -eq 0) {
    Write-Output "[FAIL] Image file is empty (0 bytes): $ImageFile"
    exit 1
}

# Expected sizes (bytes) for common sector-image types. Flux formats (.scp/.hfe)
# have variable sizes and are only checked for non-emptiness.
$expected = @{
    ".adf" = @(901120, 1802240) # Amiga DD / HD
    ".img" = @(184320, 327680, 368640, 737280, 819200,
               1228800, 1474560, 2949120) # common PC sizes
    ".ima" = @(737280, 1474560)
    ".st"  = @(368640, 409600, 737280, 819200) # Atari ST
    ".d64" = @(174848, 175531) # C64 1541 (w/o +
with error info)
}

$ext = $item.Extension.ToLower()
if ($expected.ContainsKey($ext)) {
    if ($expected[$ext] -contains $item.Length) {
        Write-Output ("[OK] Size check passed: {0:N0} bytes is valid for {1}" -f
$item.Length, $ext)
    } else {
        Write-Output ("[WARN] Unusual size for {0}: {1:N0} bytes (expected one of:
{2})" -f `
```

```

        $ext, $item.Length, ($expected[$ext] -join ", "))
    }
} else {
    Write-Output ("[INFO] No size table for {0} - skipping size check ({1:N0}
bytes)." -f $ext, $item.Length)
}

$hash = (Get-FileHash $ImageFile -Algorithm SHA256).Hash
$sidecar = "$ImageFile.sha256"
"$hash *$([IO.Path]::GetFileName($ImageFile))" | Out-File -FilePath $sidecar -
Encoding ascii
Write-Output "[OK]    SHA-256: $hash"
Write-Output "[OK]    Checksum written to $sidecar"
exit 0

```

Later, you can verify an image against its sidecar at any time: `certutil -hashfile image.adf SHA256` and compare, or use any checksum tool.

14.4 Converting flux images to sector images

`gw.exe` itself can convert between image types (`gw convert`). A classic pipeline is: read everything as raw `.scp` flux for archival, then automatically derive a usable `.adf` or `.img`:

Script: `Convert-Image.bat`

```

@echo off
REM Convert-Image.bat – converts a flux image (e.g. .scp) to a sector image
REM using "gw.exe convert".
REM
REM Usage: Convert-Image.bat "image.scp" <format> <target-extension>
REM Example arguments in the Post-Action editor:
REM     "{ImageFile}" amiga.amigados adf          -> image.adf
REM     "{ImageFile}" ibm.1440 img                -> image.img
REM
REM Adjust GW below if gw.exe is not on your PATH.
set "GW=gw.exe"

if "%~3"==" " (
    echo Usage: Convert-Image.bat image.scp format target-extension
    exit /b 2
)

echo Converting %~nx1 to %~n1.%3 (format %2) ...
"%GW%" convert --format %2 "%~1" "%~dpn1.%3"
if errorlevel 1 (
    echo [ERROR] gw convert failed with exit code %errorlevel%
    exit /b 1
)
echo Done: %~dpn1.%3
exit /b 0

```


14.5 Copying finished images to a backup location

Script: `Copy-ToBackup.ps1`

```
<# Copy-ToBackup.ps1
Copies the finished image (and its .sha256 sidecar, if present) to a backup
folder or NAS share, preserving the file name.

Post-Action setup:
  Type:      PowerShell Script
  File:      C:\...\scripts\postactions\Copy-ToBackup.ps1
  Arguments: -ImageFile "{ImageFile}" -Destination "\\NAS\FloppyArchive"
#>
param(
  [Parameter(Mandatory = $true)][string]$ImageFile,
  [Parameter(Mandatory = $true)][string]$Destination
)

$errorActionPreference = "Stop"
if (-not (Test-Path $ImageFile)) { throw "Image not found: $ImageFile" }
New-Item -ItemType Directory -Force -Path $Destination | Out-Null

Copy-Item $ImageFile -Destination $Destination -Force
Write-Output "Copied $($([IO.Path]::GetFileName($ImageFile)) -> $Destination"

$sidecar = "$ImageFile.sha256"
if (Test-Path $sidecar) {
  Copy-Item $sidecar -Destination $Destination -Force
  Write-Output "Copied checksum sidecar as well."
}
exit 0
```

14.6 A recommended action chain for archival reads

Order matters — actions run top to bottom:

#	Action	Type	Arguments
1	Validate image	PowerShell Script	-ImageFile "{ImageFile}"
2	Convert SCP → ADF	Batch Script	"{ImageFile}" amiga.amigados adf
3	Zip image	PowerShell Script	-ImageFile "{ImageFile}"
4	Copy to NAS	PowerShell Script	-ImageFile "{ImageFile}" -Destination "\\NAS\FloppyArchive"

15. Troubleshooting and FAQ

No devices are found on startup. Check the USB cable and try  **Devices** →  **Auto-Detect**. Verify in the Windows Device Manager that a COM port appears when you plug in the GreaseWeazle. If the port exists but detection fails, add the device manually by selecting its port.

"gw.exe exited with code 1" immediately after starting a job. Open  **View Log** — the first lines usually name the cause: `gw.exe` not found (fix the path in  Settings), an unknown `--format` name, a disk that is not inserted, or a drive that is not responding.

The command preview shows my option but the disk is read wrongly. Read the log: `gw.exe` prints exactly what it did. Common pitfalls: wrong *Disk Format*, missing `step=2` for 40-track disks in 80-track drives, and forgetting `--dense1 dd` with some HD drives reading DD media.

Red cells appear although the job completes. Those tracks had errors despite the retries. Try again with a higher `--retries` count, more **Revolutions**, after cleaning the disk/disk head — or accept the loss if the disk is degraded.

Post-action did not run. Post-actions only run after **successful** jobs. Also check the *En* column (enabled ✓) and the log's `=== Post-Actions ===` section for error output and exit codes.

Paths with spaces break my script. Quote the tokens in the Arguments field: `-ImageFile "{ImageFile}"`.

Language switch shows a mix of English and German. Some UI elements refresh only after an application restart, as noted in the Settings dialog.

Can I write-protect against accidental writes? Yes — physically: slide the write-protect tab on the disk. The drive hardware then blocks all writes regardless of software.

16. Glossary — Floppy and GreaseWeazle Terminology

Ordered roughly from "physical" to "logical".

Term	Explanation
Flux (magnetic flux)	The magnetisation pattern on the disk surface. Data is stored as <i>flux transitions</i> — points where the magnetic polarity flips. A GreaseWeazle records the precise timing between these transitions, which is why it works with any format.
Flux image	An image file that stores the raw flux timing (e.g. <code>.scp</code>). It preserves <i>everything</i> , including copy protection — but emulators mostly want sector images.
Sector image	An image file that stores only the decoded user data, sector by sector (e.g. <code>.img</code> , <code>.adf</code> , <code>.st</code>). Compact and emulator-friendly, but only possible when the format is known and intact.
Track	One circular ring of data on one side of the disk. In gw.exe terminology a track is identified by <i>cylinder + head</i> (e.g. <code>T12.0</code> = cylinder 12, head 0).

Term	Explanation
Cylinder	All tracks at the same head position across both sides. An 80-track disk has cylinders 0–79. GWCopyPro's <code>c=</code> range selects cylinders.
Head / Side	The read/write head; head 0 = underside, head 1 = top side of the disk. Double-sided disks use both. GWCopyPro's <code>h=</code> selects heads.
Sector	A subdivision of a track (typically 512 bytes on PC disks, 9–18 sectors per track). Sector images are organised this way.
Soft-sectored	The normal case: sector boundaries are defined by data marks; the disk has a single index hole.
Hard-sectored	Old media where each sector has its own physical index hole. Needs the <code>--hard-sectors</code> flag.
Index hole / index pulse	A small hole in the disk; a sensor generates one pulse per revolution, marking the "start" of each track.
Revolution	One full 360° turn of the disk. <code>--revs</code> selects how many revolutions of flux are captured per track — more revolutions give the decoder more chances on weak bits.
RPM	Revolutions per minute — 300 for most drives, 360 for 5.25" HD drives (and 8").
TPI (tracks per inch)	Track density: 48 tpi (40-track 5.25"), 96 tpi (80-track 5.25"), 135 tpi (3.5").
Double stepping (<code>step=2</code>)	Reading a 40-track (48 tpi) disk in an 80-track (96 tpi) drive: the drive must step <i>twice</i> per logical track.
Density (SD/DD/QD/HD/ED)	Single / Double / Quad / High / Extra-high density — generations of media with increasing capacity (e.g. 3.5": DD=720KB, HD=1.44MB, ED=2.88MB). Media and recording mode must match.
Density select (<code>densel</code>)	A signal line telling the drive which density mode to use. <code>--densel hd/dd/ed</code> overrides it — occasionally needed when reading DD disks in HD drives.
Bitrate	The data rate of the recording, e.g. 250 kbit/s (DD) or 500 kbit/s (HD). <code>--bitrate</code> can force it; 0 = auto.
FM / MFM / GCR	Encoding schemes that translate bits into flux transitions. FM (single density, oldest), MFM (most PC/Amiga/Atari formats), GCR (Commodore 1541, old Macintosh).
Flippy disk	A 5.25" disk written on both sides by <i>physically flipping it over</i> in a single-sided drive (common for C64/Apple II). Reading side B in a PC drive needs tricks: the <code>--reverse</code> flag and/or a "flippy-modified" drive with head offsets (<code>h0.off=</code> / <code>h1.off=</code>).
hswap	"Head swap" — corrects drives whose two heads are wired in reverse.

Term	Explanation
Write precompensation (--precomp)	Deliberately shifting flux transitions by fractions of a microsecond when writing inner tracks, to counteract the magnetic drift ("bit shift") that occurs there.
TG43 (--gen-tg43)	"Track Greater than 43" — a signal some 8" drives need on tracks > 43 to lower the write current.
Write-protect	The physical tab/notch on a disk. When set, the drive hardware refuses all writes.
Shugart bus	The classic 34-pin floppy interface. Drives are addressed either as 0-3 (straight cable, Shugart standard) or a/b (PC twisted cable) — this is what --drive selects.
COM port	The virtual serial port (e.g. COM3) Windows creates for the GreaseWeazle's USB connection — this is what --device selects.
Firmware	The program running <i>on</i> the GreaseWeazle board. gw.exe info reports its version; gw.exe update upgrades it.
SCP (.scp)	<i>SuperCard Pro</i> flux image format — the de-facto standard for raw flux archival.
HFE (.hfe)	Flux-level format of the HxC floppy emulator ecosystem — ideal when the target is a hardware floppy emulator (Gotek).
ADF (.adf)	<i>Amiga Disk File</i> — sector image of an AmigaDOS disk: 901,120 bytes (DD) or 1,802,240 bytes (HD).
ADZ (.adz)	A gzip-compressed ADF. Extract with unar.exe before writing.
IMG / IMA (.img)	Plain sector image, most commonly of PC/MS-DOS disks (e.g. 1,474,560 bytes for 1.44MB).
ST (.st)	Sector image of an Atari ST disk.
IPF (.ipf)	<i>Interchangeable Preservation Format</i> by the Software Preservation Society — describes protected disks precisely; GWCOPYPro supports it for writing .
D64 (.d64)	Sector image of a Commodore 1541 disk (174,848 bytes).
Preset (.gwpreset)	GWCOPYPro's own JSON file storing a complete job configuration.
Post-action	A program/script GWCOPYPro runs automatically after a successful job.

17. gw.exe Parameter Dictionary

Everything GWCOPYPro generates, explained in plain language. The first token is the **command**:

Command	Meaning
gw read <options> <image>	Read a physical disk into an image file.

Command	Meaning
<code>gw write <options> <image></code>	Write an image file onto a physical disk.

Other useful `gw.exe` commands (run them manually in a terminal for tasks GWCopyPro's UI doesn't cover): `gw info` (device/firmware info), `gw convert` (convert image ↔ image), `gw erase` (bulk-erase a disk), `gw clean` (head-cleaning cycle with a cleaning disk), `gw reset`, `gw update` (firmware update).

GWCopyPro itself also calls one of these internally, outside of a read/write job:

Command	Used by
<code>gw seek --device COMx --drive N 0</code>	The ⚡ Blink button on a device tile (5.2) and the LED-guided insert sequence in group repetitive jobs — selecting a drive lights its LED for the duration of the command, producing a visible blink when pulsed on a cycle.

There is no disk-presence check behind ✓ **Disk inserted** in the Batch Insert dialog — `gw.exe` has no reliable, dedicated way to detect whether a disk is physically in a drive, so GWCopyPro trusts your confirmation directly and advances immediately.

Parameters generated by GWCopyPro

Parameter	Used for	Plain-language meaning
<code>--device COMx</code>	read + write	Which GreaseWeazle to use, by its COM port. Omitted = auto-select.
<code>--drive a b 0-3</code>	read + write	Which floppy drive on the ribbon cable: <code>a/b</code> for PC-style twisted cables, <code>0-3</code> for straight Shugart cables.
<code>--format <name></code>	read + write	The logical disk format (e.g. <code>ibm.1440</code> , <code>amiga.amigados</code>). Tells <code>gw.exe</code> how to decode/encode sectors. Omit for raw flux work.
<code>--tracks= <spec></code>	read + write	Which tracks to process — see breakdown below.
<code>--revs N</code>	read	Revolutions of flux captured per track (default 1; more = better recovery of weak data).
<code>--densel hd dd ed</code>	read + write	Force the density-select line to high/double/extra density instead of auto.
<code>--bitrate N</code>	read + write	Force the data rate (kbit/s) instead of auto-detecting.
<code>--retries N</code>	read	Extra read attempts per track when bad sectors are found.
<code>--no-clobber</code>	read	Never overwrite tracks already present in the output image (resume support).
<code>--raw</code>	read	Store raw flux without decoding — even when a format is given.

Parameter	Used for	Plain-language meaning
<code>--reverse</code>	read + write	Reverse the track data direction — for side B of floppy disks.
<code>--hard-sectors</code>	read + write	Handle hard-sectored disks (multiple index holes per revolution).
<code>--erase</code>	write	Erase each track before writing it.
<code>--verify</code>	write	Read back and compare every written track.
<code>--precomp N</code>	write	Write precompensation in microseconds.
<code>--gen-tg43</code>	write	Generate the /TG43 signal for 8" drives.

The `--tracks=` specifier in detail

```
--tracks=c=0-79:h=0-1:step=2:hswap:h0.off=+1:h1.off=-1
```

cylinders

heads

double

head

flippy head

first-last

0/1/

step

swap

cylinder offsets

both

Component	Meaning
<code>c=A-B</code> or <code>c=N</code>	Cylinder range (inclusive) or a single cylinder.
<code>h=0-1</code> / <code>h=0</code> / <code>h=1</code>	Both heads, or only head 0 / head 1.
<code>step=N</code>	Physical head steps per logical cylinder (2 = 40-track disk in 80-track drive).
<code>hswap</code>	Swap the two physical heads.
<code>h0.off=±N</code> / <code>h1.off=±N</code>	Per-head cylinder offset for flippy-modified drives.

Removed legacy flags (pre-v0.24, *never* use with modern gw.exe): `--scyl`, `--ecyl`, `--shead`, `--ehead`, `--single-sided`. Their function is fully covered by `--tracks=`.

Worked examples

Goal	Command line (as shown in GWCOPYPro's preview)
Archive an unknown disk as raw flux, 3 revolutions	<code>gw.exe read --device COM3 --revs 3 "disk.scp"</code>
PC 1.44MB disk to IMG with retries	<code>gw.exe read --device COM3 --format ibm.1440 --retries 3 "disk.img"</code>
Amiga DD disk to ADF	<code>gw.exe read --device COM3 --format amiga.amigados "game.adf"</code>

Goal	Command line (as shown in GWCOPYPro's preview)
40-track 5.25" disk in an 80-track drive	<code>gw.exe read --device COM3 --tracks=c=0-39:h=0-1:step=2 "old.scp"</code>
Side A only (single-sided format)	<code>gw.exe read --device COM3 --tracks=c=0-79:h=0 "side_a.scp"</code>
Write an ADF back to disk, verified	<code>gw.exe write --device COM3 --format amiga.amigados --erase --verify "game.adf"</code>

18. Appendix

18.1 File and folder reference

Location	Contents
<code><app_folder>\Logs\Job_<Type>_<ID>_<timestamp>[_diskN]\gw_output.log</code>	Per-job (per-disk) log.
<code>%APPDATA%\GreaseWeazleManager\settings.json</code>	Application settings.
<code>%APPDATA%\GreaseWeazleManager\Presets*.gwpreset</code>	Saved job presets (JSON).
<code><app_folder>\tools\lsar.exe, unar.exe</code>	Bundled archive tools.
<code>scripts\postactions*</code> (repository)	The ready-to-use post-action scripts from chapter 14.

18.2 Useful links

- GreaseWeazle project & downloads: <https://github.com/keirf/greaseweazle>
- GreaseWeazle wiki (drive wiring, supported formats): <https://github.com/keirf/greaseweazle/wiki>
- The Unarchiver command-line tools (lsar/unar): <https://theunarchiver.com/command-line>
- unrar command-line: https://www.rarlab.com/rar_add.htm
- SuperCard Pro (.scp format): <https://www.cbmstuff.com/>
- Software Preservation Society (.ipf): <https://www.softpres.org/>